# Sarcasm-Aware Sentiment Analysis for Indonesian Twitter Using IndoBERT, RoBERTa, and mBERT

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***Abstract*—Sarcasm poses a persistent challenge for sentiment analysis systems in Indonesian social media, where informal language, slang, code-mixing, and creative spelling patterns further complicate automated text understanding. This study proposes a sarcasm-aware sentiment analysis framework integrating three Transformer-based sarcasm detectors IndoBERT, RoBERTa, and Multilingual BERT (mBERT) with a rule-based sentiment override mechanism applied to Indonesian X (Twitter) data. Methods: Models were trained on a class-balanced dataset derived from the w11wo/twitter_indonesia_sarcastic corpus, augmented with manually annotated binary sentiment labels. A confidence threshold of 0.8 was applied to trigger sentiment corrections for high-confidence sarcastic instances. Upsampling was used to address the severe 95:5 class imbalance in the sarcasm label. Results: Among the three sarcasm detection architectures, RoBERTa achieved the highest F1-Score (0.7716), while mBERT obtained the highest accuracy (0.9619). For baseline sentiment classification, IndoBERT achieved the strongest performance (Accuracy: 0.9646, F1-Score: 0.9635). Following sarcasm-aware enhancement, all three models showed improvement: IndoBERT increased its F1-Score from 0.9635 to 0.9656 (+0.0021), RoBERTa improved from 0.9404 to 0.9428 (+0.0025), and mBERT showed a marginal gain from 0.9262 to 0.9264 (+0.0002).***

***Keywords*—sarcasm detection, sentiment analysis, IndoBERT, RoBERTa, Multilingual BERT, Transformer models, Indonesian NLP, social media, Twitter, natural language processing***

## I. Introduction

The use of sarcasm is prevalent all over the world today, including by Indonesians. Sarcasm can often convey criticism, annoyance, or humour in an indirect way. When communicating on social media through writing, it is difficult to determine if the writer intended to be sarcastic because there are no vocal tones or facial expressions to provide guidance. This seems to be a particular problem in Indonesian online discourse, where users tend to routinely use informal forms of slang, creatively misspelled words, emojis, and code-mix between English and Indonesian. Because of these features, there is an added level of difficulty for automated text-analysis software/models in identifying if and how text is sarcastically conveyed; [1]. Moreover, much of the research on this area has also identified that many datasets related to these topics have been imbalanced and have therefore required novel approaches, including the application of data augmentation techniques to maintain model performance [2].

With the rapid growth of user-generated content on X (Twitter), sentiment analysis has become an essential tool for understanding public opinion, analyzing politics, and social trends. However, conventional sentiment analysis systems often rely on lexical polarity, making them less effective at interpreting sarcastic expressions. As a result, a sentence such as "Wow harganyaa sangat murah sekali!!!" may be classified as positive even though it is intended as criticism. Previous studies have shown that ignoring sarcasm can reduce the accuracy of sentiment classification [1], [3].

This limitation is particularly important in applications such as public opinion analysis. For example, sentiment analysis has been used to examine public responses to Indonesia's New Capital (IKN). In this context, contextual language models such as IndoBERT have been shown to outperform traditional approaches like SVM because they are better at capturing the intended meaning behind indirect or sarcastic statements [4].

Transformer-based models have become the main approach for sarcasm detection because they can better understand contextual information than traditional machine learning methods. BERT-based models, for example, have shown strong performance by capturing sentence context, which is important for recognizing sarcastic expressions [1]. Similar results were reported in the IdSarcasm benchmark, where Transformer models outperformed conventional machine learning and shallow neural network approaches. [5]

Research in Indonesian sarcasm detection has followed a similar direction. Early studies mainly relied on lexicon-based methods such as SentiStrength, but these approaches struggled to identify implicit sarcasm [6]. More recent studies have shifted to deep learning and Transformer models, which generally achieve better and more stable performance [7].

Several researchers have also proposed hybrid architectures. Models that combine multi-head attention with CNN and Bi-GRU have shown promising results for Indonesian–English code-mixed text [8]. Similar findings have been reported in other languages, where Transformer-based models consistently outperform traditional feature-based methods [9].

Even though Transformer models are widely used, comparisons between different Transformer architectures for Indonesian Twitter data are still limited. Since sarcasm in Indonesian often depends on context and informal language, this study compares IndoBERT, RoBERTa, and mBERT for sarcasm detection. It also investigates whether incorporating sarcasm detection can improve sentiment analysis for Indonesian social media.

## II. Related works

Previous studies on sentiment analysis and sarcasm detection have used a variety of methods. Early work mostly relied on traditional lexical approaches, while more recent studies have shifted toward neural network models that are more effective at handling social media language. Despite these advances, research on sarcasm detection in Indonesian posts on X is still limited. This section reviews prior work and identifies the gap that this study aims to address.

### *A. Traditional Sentiment Analysis*

Before focusing on deep learning, sarcasm detection mainly depended on lexicon-based methods. These approaches classified text by analyzing sentiment-related words, but they often failed to recognize implicit sarcasm because they did not account for context. For example, R. Kusumastuti et al. applied SentiStrength to Indonesian posts on X and reported that the method struggled to identify sarcastic expressions [6].

As research progressed, traditional machine learning models became more common. Although these methods generally improved classification performance, they still faced challenges when processing the informal and ambiguous language commonly found on social media. For instance, F. Z. Lubis and R. Kurniawan achieved 88% accuracy using Naive Bayes for sentiment analysis of discussions about the naturalization of Indonesian National Team players on X [10]. However, they also reported that the model had difficulty recognizing sarcasm because it relied mainly on word frequency rather than contextual meaning. As a result, words such as "main" ("play") could be interpreted differently depending on the context, making sarcastic expressions more difficult to detect.

To overcome these limitations, later studies began incorporating additional linguistic features instead of relying only on sentiment words. For example, V. Govindan and V. Balakrishnan used features such as hyperbole, interjections, intensifiers, punctuation, and repeated letters with Support Vector Machine and Random Forest classifiers. Their results showed that these handcrafted features improved sarcasm classification performance [11].

Researchers have also found that sarcasm is expressed differently across users. According to M. Khadafi, factors such as gender can influence how sarcasm appears in informal text, including differences in writing style, direct sarcastic statements, and emoji usage. These findings suggest that sarcasm is highly dependent on context and user characteristics, making it difficult for traditional methods to detect consistently [12].

### *B. Preprocessing and Augmentation*

The quality of the training data is another factor that influences sarcasm detection performance. Besides selecting an appropriate model, proper data preprocessing can significantly improve the effectiveness of pre-trained language models. For example, N. H. Jeremy [13] demonstrated that removing HTML tags and handling common social media text patterns, such as code-mixing and abbreviations, resulted in better model performance on Indonesian data.

Class imbalance is another challenge in sarcasm detection because sarcastic posts are much less common than non-sarcastic ones. As a result, models may become biased toward the majority class and struggle to recognize sarcasm. To reduce

this problem, A. T. Handoyo and Hidayaturrahman applied Global Vector (GloVe) representations to improve the balance of the training data. Their findings showed that a more balanced dataset can help improve sarcasm detection performance [2].

*C. Deep Learning and Transformers*

Sarcasm detection has advanced considerably with the development of deep learning models. Earlier methods mainly depended on individual words and often struggled to understand the intended meaning of a sentence. More recent studies have shown that considering the surrounding context leads to better performance. For example, H. Yaghoobian, H. R. Arabnia, and K. Rasheed reported that models using conversational context outperform those that analyze sentences independently [14]. Likewise, a review by S. O. Slim, A. E. Aboutabl, and A. D. Yacoub, which examined 40 previous studies, found that Transformer-based models with attention mechanisms consistently achieve stronger results than traditional approaches. Together, these findings demonstrate the value of context-aware models for sarcasm detection [15].

The introduction of pre-trained language models has also improved sarcasm detection. Because these models are trained on large collections of text, they are better at understanding the context and relationships between words. C. I. Eke et al. showed that adding contextual features to BERT improves sarcasm detection performance [1]. Similarly, P. Parameswaran et al. found that BERT effectively captures semantic relationships that help identify sarcastic expressions [3]. In another study, S. Kaya and B. Alatas reported that BERT achieved higher performance than a CNN+BiLSTM model, reaching an accuracy of 94.39% and an F1-score of 93.83% [16].

Besides improving model architectures, some researchers have focused on prompting techniques to enhance sarcasm detection. For example, M. Iskandardinata, W. Christian, and D. Suhartono proposed Pragmatic Metacognitive Prompting as a way to improve model understanding. Although the method led to some performance improvement, the gains were relatively small. Their findings also showed that conventional Transformer models continue to provide strong and reliable performance [17].

Different RoBERTa architectures have also been tested for sarcasm detection. S. Pawestri, Murinto, and M. Auzan found that RoBERTa-CNN outperformed RoBERTa-RNN, achieving an accuracy of 89.1%. The CNN-based model was also faster, making it a more efficient choice for this task [18].

*D. Sarcasm Detection in the Indonesian Context*

In Indonesian NLP, localized Transformer models usually perform better than traditional methods. For example, Riansyahputra et al. compared TF-IDF with SVM against IndoBERT for analyzing discussions about Indonesia's New Capital (IKN). They found that SVM performed worse than IndoBERT, showing that contextual models work better. In another study, Sabrina, Shiddieq, and Roji analyzed 5,637 Indonesian tweets about the boycott of Israeli products and also found that IndoBERT beat SVM in detecting sarcastic sentiment (92.20% vs 86.15%), even though SVM was faster in terms of computation [19].

Additional research using the IdSarcasm dataset indicates that Transformer models surpass traditional methods [5]. This supports the idea that context is very important when understanding sarcasm in Indonesian text.

When people talk online in Indonesian they use language in ways. To understand this scholars tried using computer models that combine different techniques [8]. Some people, like Wajdi, Purbolaksono and Adiwijaya made a model that uses IndoBERT and something called a Gated Recurrent Unit or GRU for short. They used this model to look at a lot of texts from Indonesian Twitter and Instagram. 14,000 Texts that people had labeled by hand. Their model used IndoBERT to understand the meaning of words in context and the GRU to see how the words were connected. This worked well.

Their model was right 76.88 percent of the time which is better than using just IndoBERT, which was right 74.75 percent of the time. The Indonesian text, from Twitter and Instagram was analyzed using this model. The model that combines IndoBERT and GRU is a way to analyze Indonesian text especially when people are being sarcastic [20].

*E. Comparison, Discussion, and Research Gap*

Reading people's writings on this topic reveals a common point: traditional methods can indicate emotions, yet we require modern approaches to discern when individuals are humorous or sarcastic. Traditional methods can reveal much about individuals, including their preferences and dislikes. In Indonesia, there are computer programs such as IndoBERT that excel at comprehending the language. These programs are extremely beneficial.

Merging these programs can assist us in understanding conversations that are difficult to grasp, such as when individuals discuss topics humorously or express sarcasm. Computer programs such as IndoBERT excel at comprehending the language in Indonesia [4], [19], [20].

There is still a big problem that we do not know the answer to. Even though people have tested IndoBERT on its own or with programs, nobody has compared all the best new programs to see which one is the best at understanding Indonesian Twitter comments. We do not know how well IndoBERT, which is made for Indonesia, can detect sarcasm when compared to good general programs like RoBERTa or programs that can understand many languages like Multilingual BERT, especially when people use a lot of slang and informal language. This study is trying to answer this question by comparing all these programs to see which one is the best for Twitter [18] [20]. Indonesian Twitter comments are very hard to understand because they use a lot of sarcasm and slang. We need to find out which program is the best, at understanding Indonesian Twitter comments.

## III. METHODOLOGY

### A. Data

This study used a collection of texts from social media, which were taken from Twitter. The study mainly used the w11wo/twitter_indonesia_sarcastic dataset to find sarcasm, which is a dataset that has labels to help detect sarcasm in texts. Each line in the dataset has the text and a label that says if the text is sarcastic or not, with 0 meaning the text is not sarcastic and 1 meaning the text is sarcastic.

Since the original dataset does not include sentiment annotations, an additional labeling process was conducted to assign sentiment labels to each data instance. The sentiment labeling was performed manually, where each tweet was categorized into binary sentiment classes: 0 for negative sentiment and 1 for positive or neutral sentiment. This additional annotation enables the construction of a dual-label dataset suitable for both sarcasm detection and sentiment analysis tasks.

The dataset contains tweet text, sarcasm labels, and sentiment labels. However, it is highly imbalanced, with about 95% non-sarcastic tweets and only 5% sarcastic ones. This creates a problem during training, since a model could predict all tweets as non-sarcastic and still reach around 95% accuracy, even without detecting any sarcasm.

### B. Preprocessing

The dataset was divided into training and testing sets using an 80:20 ratio with stratified sampling to preserve the original label distribution. This approach ensures that the model is evaluated under realistic conditions, where sarcastic tweets remain relatively rare.

Because the training data was highly imbalanced, sarcastic samples were duplicated until both classes were balanced. This upsampling process was applied only to the training set, while the test set retained its original distribution (approximately 95% non-sarcastic and 5% sarcastic) to provide a realistic evaluation.

Before training the model, each tweet was preprocessed and tokenized using the tokenizer corresponding to each model (IndoBERT, RoBERTa, or mBERT). The tokenized sequences were then standardized to a maximum length of 128 tokens. Longer sequences were truncated, whereas shorter ones were padded to ensure consistent input across all models.

### C. Modeling

The proposed framework consists of two stages: sentiment analysis and sarcasm detection. IndoBERT was first fine-tuned for sentiment classification to generate the initial sentiment prediction.

For sarcasm detection, three Transformer models were evaluated: IndoBERT, RoBERTa, and mBERT. These models represent both language-specific and multilingual approaches, allowing their performance on Indonesian text to be compared under the same experimental setting.

All models were trained using the same hyperparameters. Each model was optimized with the AdamW optimizer, a weight decay of 0.01, a batch size of 16, and three training epochs. Using identical training settings helps ensure that any performance differences are mainly due to the model architecture rather than differences in the training process.

To provide a clearer overview of the proposed sarcasm-aware sentiment analysis framework, the overall architecture is illustrated in Figure 1.

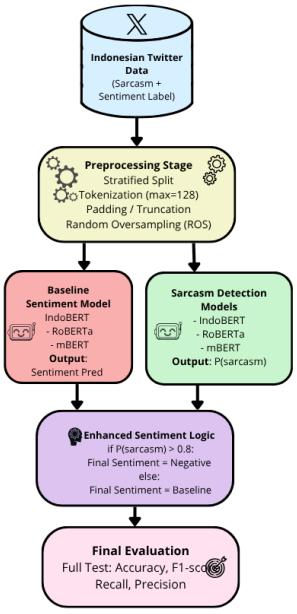



Fig. 1. Sarcasm-aware enhanced sentiment analysis pipeline

As illustrated in Figure 1, the proposed framework performs sentiment analysis and sarcasm detection simultaneously using IndoBERT, RoBERTa, and mBERT. The sarcasm score is then used in a rule-based decision step, where the original sentiment prediction is modified only if the predicted sarcasm probability is greater than 0.8.

*D. Enhanced Sentiment Mechanism*

The main idea of this study is to combine sarcasm detection with sentiment analysis to improve the final sentiment prediction.

$y_{sent}$ used to represent the sentiment predicted by the baseline model, while $P_{sarc}$ represents the probability that the text is sarcastic. The final sentiment prediction, $y_{final}$, is then determined using the following rule:

$$y_{final} = \begin{cases} 0, if\ P_{sarc} > 0.8 \\ y_{sent}\ otherwise \end{cases} \quad (1)$$

The sentiment prediction is updated only when the model identifies sarcasm with a confidence score above 0.8. This threshold reduces unnecessary corrections while allowing the framework to capture tweets that are likely to be sarcastic. The rule is based on the observation that sarcastic expressions often contain positive words but convey negative sentiment.

The process starts by sending the text to the sentiment analysis model, which gives an initial sentiment prediction. Next, the same text is analyzed by the sarcasm detection model. The predictions from both models are then used to decide the final sentiment. This helps the framework handle sarcastic text better and improves the overall sentiment classification results.

*E. Evaluation*

All models are evaluated on the same full test dataset using Accuracy, Precision, Recall, and F1-score. Because the dataset is highly imbalanced, with most samples being non-sarcastic, accuracy alone is not enough to judge performance. A model that always predicts the majority class can still achieve high accuracy while failing to detect sarcasm.

## IV. Result and Discussion

*A. Sarcasm Detection Performance*

As the result shown in Table I, RoBERTa achieved the highest F1-score (0.7716), with an accuracy of 0.9556, precision of 0.8065, and recall of 0.7108. mBERT followed with a slightly lower F1-score (0.7567) but the highest accuracy (0.9619), along with a precision of 0.7851 and recall of 0.7315. IndoBERT recorded an F1-score of 0.7293, accuracy of 0.9430, precision of 0.7108, and recall of 0.7486.

TABLE I. Sarcasm Detection Matrix per Model

| Model | IndoBERT | RoBERTa | mBERT |
|---|---|---|---|
| Accuracy | 0.9557 | 0.9522 | 0.9619 |
| Precision | 0.7919 | 0.7573 | 0.8528 |
| Recall | 0.6910 | 0.7879 | 0.7049 |
| F1-Score | 0.7293 | 0.7716 | 0.7567 |

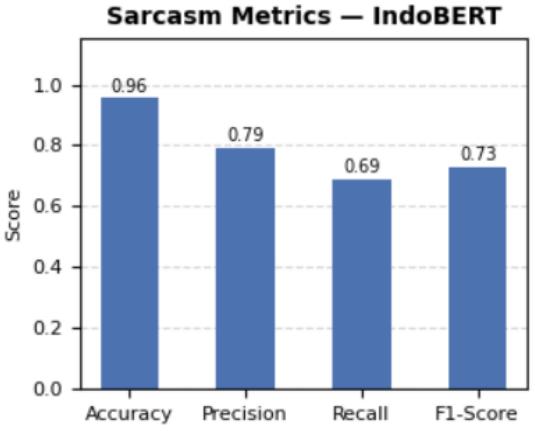



Fig. 2. IndoBERT Bar Chart Sarcasm Metric

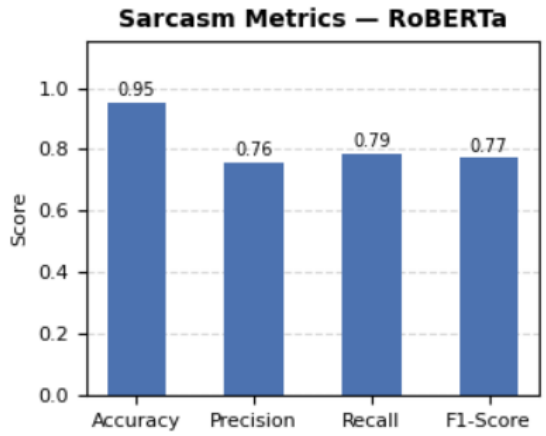



Fig. 3. RoBERTa Bar Chart Sarcasm Metric

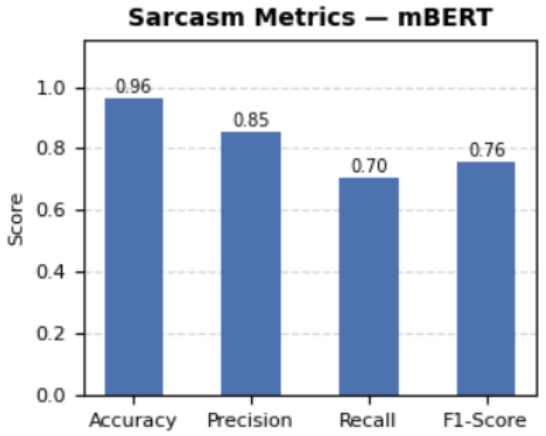



Fig. 4. mBERT Bar Chart Sarcasm Metric

The original dataset contained only about 5% sarcastic samples, making sarcasm detection more difficult. We addressed this problem by balancing the training data through upsampling. After training, all three models were able to detect sarcasm in Indonesian tweets, although their performance varied. The differences in performance may be related to the different pre-training data and architectures of the models.

*B. Sentiment Analysis on Baseline vs Enhanced*

Table II compares the baseline sentiment classification results of the three models. Among the three models, IndoBERT achieved the strongest overall performance, recording an accuracy of 0.9646 and an F1-score of 0.9635. It also obtained the highest precision (0.9542) and recall (0.9728). RoBERTa ranked second with an accuracy of 0.9578 and an F1-score of 0.9404, followed by mBERT, which achieved an accuracy of 0.9520 and an F1-score of 0.9262. Both models also maintained high recall values of 0.9631 and 0.9601, respectively, although their precision scores were lower than IndoBERT at 0.9187 and 0.8941.

TABLE II. Sentiment Analysis: Baseline vs Enhanced (Macro Average)

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| IndoBERT (Baseline) | 0.9646 | 0.9671 | 0.9607 | 0.9635 |
| RoBERTa (Baseline) | 0.9421 | 0.9425 | 0.9386 | 0.9404 |
| mBERT (Baseline) | 0.9277 | 0.9246 | 0.9283 | 0.9262 |
| IndoBERT (Enhanced) | 0.9666 | 0.9669 | 0.9645 | 0.9656 |
| RoBERTa (Enhanced) | 0.9440 | 0.9414 | 0.9446 | 0.9428 |
| mBERT (Enhanced) | 0.9264 | 0.9243 | 0.9295 | 0.9264 |

The sarcasm-aware enhancement improved the performance of all three models, although the amount of improvement varied. IndoBERT benefited the most, with its F1-score increasing from 0.9635 to 0.9656 (+0.22%), while recall improved to 0.9747 and precision remained unchanged. RoBERTa also showed a noticeable improvement, achieving a 0.27% increase in F1-score along with slightly higher precision and recall. In contrast, mBERT experienced only a marginal gain, with its F1-score increasing by just 0.02% and only minor changes in the other evaluation metrics.

*C. Model Architecture and Sarcasm Detection*

Each model performed well in different areas. RoBERTa achieved the highest F1-score for sarcasm detection (0.7716), while IndoBERT achieved the best baseline sentiment classification results with an accuracy of 0.9646 and an F1-score of 0.9635. mBERT recorded the highest accuracy in sarcasm detection (0.9619), but its lower F1-score suggests that it had more difficulty identifying sarcastic samples. These differences may be related to the different pre-training data and model designs used by each model.

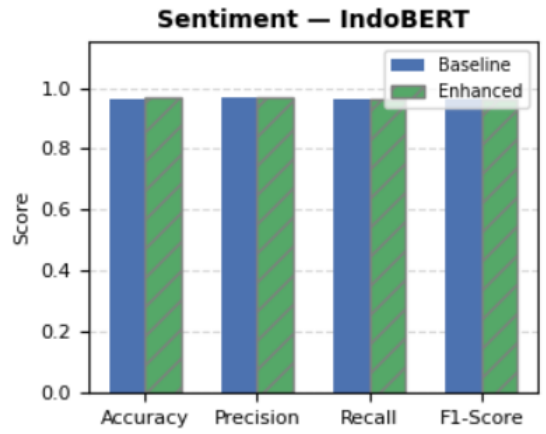



Fig. 5. IndoBERT Bar Chart Sentiment Performance on Baseline and Enhanced

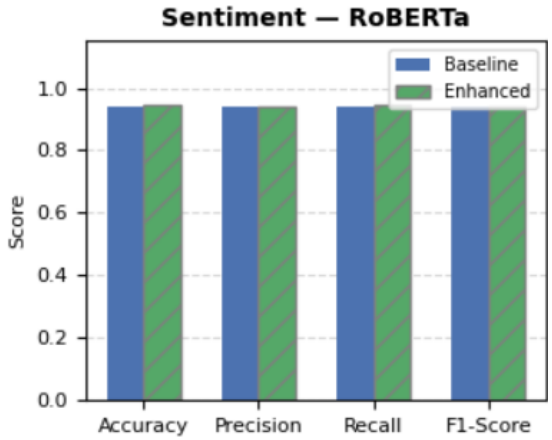



Fig. 6. RoBERTa Bar Chart Sentiment Performance on Baseline and Enhanced

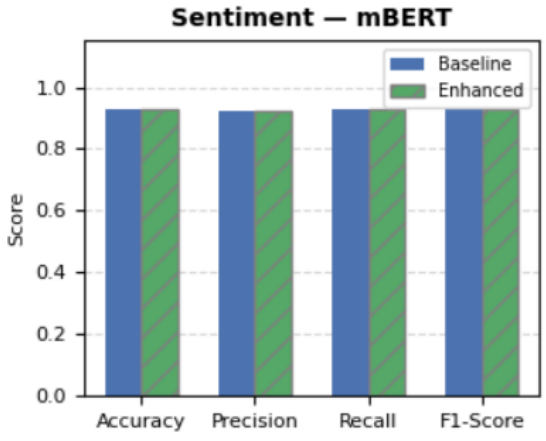



Fig. 7. mBERT Bar Chart Sentiment Performance on Baseline and Enhanced

The upsampling step helped solve the class imbalance in the training data, where only about 5% of the samples were sarcastic. As a result, all three models learned to detect sarcasm more effectively. We also applied a confidence threshold of 0.8, so the system only changed the sentiment prediction when the sarcasm score was high. If the confidence was lower than the threshold, the original sentiment prediction was kept.

*D. The Sarcasm-Aware Enhancement Mechanism*

One advantage of this framework is its modular design. The proposed framework is its modular design, which allows the sarcasm detection and sentiment analysis components to be updated independently. The confidence threshold can also be adjusted to suit different applications, while the rule-based enhancement makes the prediction process easier to understand and interpret.

The experimental results indicate that incorporating sarcasm detection improves sentiment classification, particularly for texts where positive words convey negative meaning. By considering the sarcasm prediction before producing the final sentiment label, the framework reduces these types of misclassifications. This improvement was observed across all three Transformer models, suggesting that the approach is applicable to different model architectures.

Although the performance gains were modest, the enhancement consistently improved the results. IndoBERT achieved the highest overall F1-score (0.9656), while RoBERTa showed the largest increase in F1-score. mBERT also benefited from the proposed framework, although the improvement was relatively small.

*E. Comparison with Prior Work*

The findings are consistent with previous studies showing that Transformer models outperform traditional machine learning methods for sarcasm detection. By evaluating IndoBERT, RoBERTa, and mBERT within the same framework, this study provides a direct comparison of their performance on Indonesian social media text.

Another contribution of this work is the proposed sarcasm-aware enhancement, which refines the original sentiment prediction instead of replacing the sentiment model. This allows the framework to better handle sarcastic expressions while maintaining performance on non-sarcastic text. In addition, the use of the w11wo/twitter_indonesia_sarcastic dataset, which includes both sarcasm and manually annotated sentiment labels, enables a more comprehensive evaluation.

### *F. Limitations and Future Work*

There are still several limitations in this study. Since the framework was tested only on Indonesian Twitter data, its performance on other languages or social media platforms is still unknown. The confidence threshold of 0.8 was chosen based on experimental observations instead of a systematic tuning process. In addition, this study did not include an ablation study, so the effect of each component could not be evaluated separately. Although all three models achieved better F1-scores after applying the proposed framework, the improvements were relatively small. As a result, the overall impact of the framework in real-world applications may be limited.

This study can be improved in several ways. One possible improvement is to use a validation set to determine a more suitable confidence threshold instead of relying on experimental observations. It would also be useful to conduct an ablation study to better understand how each component affects the overall performance. Future work could also explore joint fine-tuning, where the sarcasm detection and sentiment analysis models are trained together rather than separately. In addition, evaluating the framework on different languages, domains, and social media platforms would provide a better understanding of how well it generalizes to other types of data.

## V. Conclusion

A rule-based framework was developed in this study to integrate sarcasm detection into sentiment analysis for Indonesian social media text. The framework was evaluated using three Transformer models: IndoBERT, RoBERTa, and mBERT.

The comparison results show that each model has its own strengths. RoBERTa achieved the highest F1-score for sarcasm detection, while mBERT recorded the best accuracy. For baseline sentiment classification, IndoBERT delivered the strongest overall performance. After applying the sarcasm-aware enhancement, all three models showed improvements in sentiment classification. Although the improvements were not large, the results suggest that incorporating a simple rule-based sarcasm detection step can help reduce misclassification caused by sarcastic expressions.

In this framework, the sentiment prediction is updated only when the sarcasm score is higher than the confidence threshold of 0.8. This reduces unnecessary changes to non-sarcastic texts while allowing the system to better handle tweets that are likely to be sarcastic. Another advantage of the proposed framework is its modular design, which allows the sarcasm detection and sentiment analysis components to be modified or improved independently without changing the entire system.

Adding sarcasm detection can improve the reliability of sentiment analysis for Indonesian social media text. The framework produced consistent improvements across all three Transformer models, showing that sarcasm is an important factor to consider. These findings suggest that sentiment analysis systems can achieve better performance when they account for implicit meaning instead of relying only on the literal meaning of the text.

The findings from this study provide a foundation for further research. Building on this work, future studies can investigate alternative enhancement methods, compare different integration strategies, and assess the framework using data from a wider range of languages and social media platforms.


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